Suspicious Transaction Detection Report

1. Objectives

The objective of the project is to build a model to detect unusual transactions in employee credit card data, to support the audit and risk management department in identifying cases of potential fraud or violations.

2. Implementation process

Data: 742,191 transactions, including important fields: amount (MERCHANDISE_AMT), transaction date (TRANS_DT), merchant, category, division/department.

Feature engineering:

- Log transformation of amount (log_amount).
- Time information (TX_MONTH, DAY_OF_WEEK, IS_WEEKEND).
- Merchant & category rarity (rarity).
- Z-score by division/department (compared with group spending habits).

Anomaly detection method:

- Univariate: IQR, Z-score.
- Multivariate: Isolation Forest, Local Outlier Factor (LOF).
- Standardize & aggregate scores: use MinMaxScaler to return to [0,1], then calculate anomaly_score.
- Select threshold: take the top 1% of transactions with the highest anomaly_score (~7,400 transactions).

3. Analysis results

The system detects the top 1% of suspicious transactions.

Anomalous factors include:

- The amount is far above the average of the division/department.
- Merchant/category is very rare in the data.

- Transactions occur at unusual times (e.g. weekends).

Example of top 5 suspicious transactions (extracted from output):

amount	log amount	tx_month	day of week	is weekend	merchant rarity	category rarity	anomaly score	rank
379505.6	12.84663	12	6	1	0.386338	0.305173	0.75	1
368335.1	12.81675	12	4	0	0.348968	0.213014	0.733327	2
340257.2	12.73746	9	4	0	0.386338	0.305173	0.725516	3
217515	12.29003	1	4	0	0.409097	0.394419	0.689222	4
211682	12.26285	1	3	0	0.307759	0.266703	0.682098	5

4. Conclusion & recommendations

The above transactions need to be manually investigated by the audit department.

The model can be improved by:

- Adding location features (city/state/country).
- Consider adding a merchant/employee blacklist.
- Apply ensemble techniques to combine multiple detectors.

This model should be deployed periodically (quarterly/monthly) for continuous monitoring.